

Chapter 31

Sexual differentiation of the human hypothalamus: Relationship to gender identity and sexual orientation

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Abstract

Gender identity (an individual's perception of being male or female) and sexual orientation (heterosexuality, homosexuality, or bisexuality) are programmed into our brain during early development. During the intra-uterine period in the second half of pregnancy, a testosterone surge masculinizes the fetal male brain. If such a testosterone surge does not occur, this will result in a feminine brain. As sexual differentiation of the brain takes place at a much later stage in development than sexual differentiation of the genitals, these two processes can be influenced independently of each other and can result in gender dysphoria. Nature produces a great variability for all aspects of sexual differentiation of the brain. Mechanisms involved in sexual differentiation of the brain include hormones, genetics, epigenetics, endocrine disruptors, immune response, and self-organization. Furthermore, structural and functional differences in the hypothalamus relating to gender dysphoria and sexual orientation are described in this review. All the genetic, postmortem, and in vivo scanning observations support the neurobiological theory about the origin of gender dysphoria, i.e., it is the sizes of brain structures, the neuron numbers, the molecular composition, functions, and connectivity of brain structures that determine our gender identity or sexual orientation. There is no evidence that one's postnatal social environment plays a crucial role in the development of gender identity or sexual orientation.

INTRODUCTION: PROGRAMMING OF THE BRAIN

Nothing seems simpler than being able to tell whether a newborn baby is a boy or girl. After all, gender is generally determined from the appearance of our genitals at the moment of birth. The mechanism is also seemingly simple, in principle. Two XX chromosomes will become a girl, one X and one Y chromosome a boy. The boy's

Y chromosome starts the process that causes the male hormone testosterone to be produced. The presence or absence of testosterone makes the child develop male or female sex organs between the sixth and twelfth week of pregnancy. The brain differentiates along male or female lines in the second half of pregnancy, due to a male baby producing a peak of testosterone and a female baby not doing so. It is at this time that the feeling of being a

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man or a woman—our gender identity and our sexual orientation—heterosexual, homosexual, or bisexual—is programmed into the hardware of our brains for the rest of our life. Structural and functional sex differences are present in many brain areas (Bao and Swaab, 2011). During pregnancy there is already a sex-biased gene expression in a number of brain areas (Shi et al., 2016). More than one-third of the genes encoded on the Y-chromosomes have shown to be expressed in 12 brain regions of mid-gestational male fetuses (Reinius and Jazin, 2009).

That our gender identity is determined as early as in the womb has only been discovered fairly recently. Up to the 1980s it was thought that a child was born as a blank slate and that its behavior was then made male or female by social influences. In the 1960s and 1970s, this had huge consequences for the treatment of newborns with indeterminate sex organs. It was thought that it did not matter what sex one selected for one's child, as long as the sex determining operation took place soon after birth. The child's social surroundings would then ensure that his/her gender identity will get adapted to the sex organ. Since that time, however, case reports and patient associations have revealed that many lives were ruined as the result of sex assignments by means of an early operation on the basis of the anatomy of the sex organs.

A clear example of the concept of the 1960s and 1970s was J. Money's statement (1975) that "Gender identity is sufficiently incompletely differentiated at birth as to permit successful assignment of a genetic male as a girl. Gender identity then differentiates in keeping with the experiences of rearing." This sexologist from Philadelphia also believed that gender imprinting does not start until the age of 1 year, and that its development is well advanced by the age of 3–4 years (Voll, 1972; Money, 1975). His view had devastating results in the case of David Reimer, i.e., the John–Joan–John case, in which, based upon Money's view, an 8-month-old boy, who lost his penis due to a mistake during a minor surgical procedure (correcting a phimosis), was made successfully into a girl. The child's testicles were removed before he reached the age of 17 months in order to facilitate feminization. He was dressed in girl's clothes, received psychological counseling, and was given dolls as toys and estrogens in puberty (Colapinto and Marek, 2001). According to Money, the case was a great success: the child would have developed as a normal female. When I (DFS) remarked during a seminar in 1995 in Minot, in the United States, that the John–Joan–John case was the only example I knew that showed that a child's gender identity could be changed by its social environment after birth, Dr. Milton Diamond, a renowned sexologist, stood up and said that Money's claim was completely unfounded. Diamond was acquainted with Joan and knew that Joan had had his sex change reversed as an

adult and had married a woman and adopted children. Diamond made these findings public. It appeared that Reimer never identified himself as a female, and that he in fact resumed his life as a male when he was 14 years old (Diamond and Sigmundson, 1997). Years of severe depression, financial instability, and a dissolving marriage tragically led to Reimer's suicide in 2004. This story illustrates the strength of the irreversible programming influence during the intrauterine period on gender identity. Removing this child's penis and testicles, giving him psychological counseling, girls' toys and clothes, and administering estrogen during puberty could not change his gender identity.

There are also other cases describing the results of enzymatic disorders or of cloacal exstrophy that support the notion of early permanent programming of gender identity in the brain by biological factors, such as intrauterine androgen exposure, rather than by social environment and learning (Swaab, 2004; Swaab and Garcia-Falgueras, 2009) (see Table 31.1). In fact, all current data indicate that gender identity and sexual orientation arise in the womb.

With the release of DSM-V in 2013, the diagnostic label "gender identity disorder" was changed to "gender dysphoria" because of the social stigmatization of the word "disorder." In 2019, the ICD-11 replaced the term "transsexualism" with "gender incongruence" and removed it from the list of mental disorders. Rather, it is now known as a condition related to sexual health (Fernández Rodríguez, 2018).

GENDER-BASED DIFFERENCES IN BEHAVIOR

Gender-based differences in the brain and behavior are also found without a direct connection with reproductive system. One of the stereotypical differences in behavior between boys and girls, often said to be socially conditioned, is the way in which they play. Little boys are wilder and more active, and from 3 to 8 months of age prefer toy cars and play that they are soldiers, while girls prefer playing with dolls (Alexander et al., 2009). Based upon some strong doubts about the social conditioning theory after observing animals' behaviors, the author (DFS) and his wife did an experiment in the 1970s: they systematically offered their children (a girl and a boy) both kinds of toys; they found that the two children were both very consistent in making stereotypical choices, i.e., the daughter played only with dolls, while the son was interested only in toy cars.

Of course, a firm conclusion could not be drawn from observations based on the sample of two children. That this difference has a biological basis was subsequently proven by Alexander and Hines et al. (2002), who offered dolls, toy cars, and balls to vervet monkeys in

Table 31.1

Prenatal factors that influence gender identity that may result in gender dysphoria

Factor	Research subject	References
Hormones	Cloacal exstrophy 5 α -reductase-2 or 17 β -hydroxy-steroid-dehydrogenase-3 deficiency Girls with CAH	Reiner and Gearhart (2004); Meyer-Bahlburg (2005) Imperato-Mcginley et al. (1979); Wilson et al. (1993); Cohen-Kettenis (2005) Meyer-Bahlburg et al. (1996); Zucker et al. (1996); Dessens et al. (2005); de Jesus et al. (2019); Grassi et al. (2019)
	Complete androgen insensitivity syndrome results in XY heterosexual females with female gender identity	Wisniewski et al. (2000)
Genetics	Twin and family studies	Coolidge et al. (2002); Hare et al. (2009); Gómez-Gil et al. (2011); Sajeveits et al. (2015)
	Rare chromosomal disorders Chromosomal abnormalities, polymorphisms, and functional variants of genes and alleles relating to sex hormones	Hengstschläger et al. (2003) Henningsson et al. (2005); Bentz et al. (2008); Ngun et al. (2011); Fernández et al. (2014, 2018); D'Andrea et al. (2020)
Endocrine disruptors	Phenobarbital/diphantoin taken by pregnant mother DES sons	Dessens et al. (1999) Kerlin (2005)
Immunological mechanism	Fraternal birth order effect	Gómez-Gil et al. (2011); Bogaert et al. (2017)
Social factors?	No evidence	Diamond and Sigmundson (1997); Cohen-Kettenis et al. (1998); Colapinto and Marek (2001); Swaab (2003)

Note: abbreviations: CAH, congenital adrenal hyperplasia; DES, diethylstilbestrol.

the zoo (Alexander and Hines, 2002). The female monkeys preferred the dolls, whose “genital’s” they sniffed in a display of typical motherly behavior, while the male monkeys were much more interested in playing with toy cars and a ball. So toy preference is not forced on us by society, it is programmed in our brains in the womb in order to prepare us for our roles in later life, namely motherhood in the case of girls and fighting and more technical tasks in the case of boys. The gender difference in the choice of toys by monkeys shows that its underlying mechanism goes back tens of millions of years in our evolutionary history and originated before the hominids.

A peak in testosterone produced by boys in the womb appears to be responsible for this sex difference. Girls with congenital adrenal hyperplasia (CAH), resulting in that they were exposed to high testosterone levels in the womb, prefer to play with boys and with boys’ toys. They are unusually boisterous, and are often labeled “tomboy” (Nordenström et al., 2002; Mathews et al., 2009), which is a strong indication of the crucial role of testosterone levels play during pregnancy with regard to the development of such sex differences in behaviors. The opposite is the case of complete androgen insensitivity syndrome (CAIS). Due to a mutation in the androgen receptor (AR) gene, which makes the receptor insensitive

to testosterone, males with this syndrome develop into heterosexual females, in spite of the presence of high levels of testosterone (see later).

There are also clear gender-based differences in the spontaneous drawings that children make. Not only the subject matter but also the colors and compositions of boys’ and girls’ drawings differ in ways that are influenced by hormones in the womb. Girls prefer to draw human figures, especially girls and women, as well as flowers and butterflies. They use bright colors like red, orange, and yellow, and the compositions are peaceful, with the figures often standing on the same line. Boys, by contrast, prefer to draw mechanical objects, guns, conflict scenes, and vehicles like cars, trains, and planes. They often adopt a bird’s-eye view perspective and favor dark and cool colors like blue. Drawings done by 5- and 6-year-old girls with CAH resemble those of little boys, despite their being treated for their condition immediately after birth (Iijima et al., 2001). Apparently, fetal exposure to higher levels of male hormones has lasting effects on playing behavior and artistic expression. It was found that atypical toy preference does, however, not necessarily prognosticate gender dysphoria in adulthood. Rather, it is predictive of homosexuality (Wallien and Cohen-Kettenis, 2008).

Some gender-based differences in our behavior emerge so early on that they can only have arisen in the womb. As early as the first day after birth, girl babies prefer to look at faces, while boy babies prefer to look at mechanical moving objects (Connellan et al., 2000). At 1 year of age, girls already make more eye contact than boys, while girls exposed to too much testosterone in the womb make less eye contact later in childhood (Baron-Cohen et al., 2005). So here, too, testosterone in the womb plays a key role in the sex difference in social behavior.

MECHANISMS INVOLVED IN SEXUAL DIFFERENTIATION OF THE BRAIN

Brain development in general and sexual differentiation of the brain in particular are influenced by many factors that increase the variability between subjects also with respect to gender identity and sexual orientation. A number of factors have been found to increase the chance of becoming transgender (see Table 31.1) or nonheterosexual.

Hormones

The fetal gonads develop under the influence of a cascade of genes, which in boys begins with the sex-determining gene on the Y chromosome (*SRY*) (Ngun et al., 2011). The production of *testosterone* and the subsequent peripheral conversion of testosterone into *dihydrotestosterone* between weeks 6 and 12 of gestation is essential for the formation of a boy's penis, prostate, and scrotum. In contrast, the development of the female sexual organs in the womb is primarily based upon the absence of androgens. Once the differentiation of these sexual organs is settled, sexual differentiation of the brain starts in the second half of gestation, by permanent *organizing effects* of testosterone on the developing brain. The development of the brain into the female direction is based upon the absence of androgens during this period. During puberty, the brain circuits that have been organized in the womb will be *activated* by sex hormones (Swaab, 2004).

The mid-pregnancy and the first 3 months after birth are two critical periods for human brain development when testosterone levels are known to be higher in boys than in girls (De Zegher et al., 1992; Hines et al., 2002; Swaab, 2004; Van De Beek et al., 2009; Wu et al., 2009). These fetal and neonatal peaks of testosterone, together with functional changes in steroid receptors, are thought to program to a major degree the development of a boy's brain structures for the rest of his life. As sexual differentiation of the genitals takes places much earlier in development (i.e., in the 6th–12th week of pregnancy) than sexual differentiation of the brain (the second half of pregnancy), these two processes may be influenced independent from each other and can result in gender

dysphoria, i.e., a person with male sex organs who nevertheless has a female identity, or vice versa.

The fetal peak of testosterone is thought to be crucial for gender identity and sexual orientation. Gender dysphoria may be induced by abnormal fetal hormone levels. In the first place, there is an increased risk of gender dysphoria in the CAH girl cases. It should be noted that although the likelihood of gender dysphoria developing in CAH cases is 100 to 300 times higher than in the entire population (1:25,000), the risk for gender dysphoria in CAH is still <1%, whereas the probability of serious gender problems in this group is 5%. The consensus is, therefore, that girls with CAH should be raised as girls, even if their genitals are masculinized (Hughes et al., 2006). Interestingly, in a recent case, CAH even led to gender *fluctuations* during a woman's life, induced by changes in hormone levels during puberty and menopause (Grassi et al., 2019). Furthermore, approximately 15%–30% of women with CAH are nonheterosexual (displaying increased feelings of sexual attraction towards women), which also suggests a link between CAH and female sexual orientation (Zucker et al., 1996; Meyer-Bahlburg et al., 2008; Frisén et al., 2009; Wang et al., 2019).

The main mechanism responsible for gender identity and sexual orientation involves a direct effect of testosterone on the developing human brain, as is apparent from CAIS. People with this condition produce testosterone, but their bodies and brains are insensitive to it as a result of mutations in the AR gene, which leads to feminization of the external sex organs and the brain. Even if they are genetically male (XY), they become generally heterosexual women with heterosexual fantasies without gender problems (Wisniewski et al., 2000). Therefore, the level of testosterone and the sensitivity of ARs may be crucial factors involved in the variability of sexual differentiation of the brain.

Circulating testosterone is converted into estrogen in the brain by aromatase. In rodents, sexual dimorphism in the numbers and projections of aromatase-expressing neurons was found to cause masculinization of these cells independent of ARs. It was possible to induce masculinization in female rodents by either testosterone or estrogen, which indicates an important role of aromatization of androgen into estrogen in sexual differentiation in rodents (Döhler et al., 1984). It should be noted that aromatization does not play such a role in sexual differentiation of the human brain, as is shown by the phenotype of CAIS subjects. Affected XY-males develop generally as phenotypical women and experience heterosexual orientation without gender problems (Wisniewski et al., 2000). Since aromatase and estrogen receptors (ERs) are intact and estrogen levels are relatively high in XY-male-CAIS, male gender identity and

male heterosexuality are apparently not induced by estrogens following aromatization but by a direct effect of testosterone on the developing brain cells. This means that unlike in rodents, aromatization of testosterone does not play a role in these processes in human brain development.

When a male fetus has a deficiency in 5 α -reductase-2 or 17 β -hydroxy-steroid dehydrogenase-3, which prevents peripheral testosterone from being transformed into *dihydrotestosterone*, a girl-like XY child with a large clitoris will be born. These XY-children are usually initially raised as girls. However, during puberty, with the increase of their testosterone production, their “clitoris” may grow to penis size, their testicles descend, and their build begins to masculinize. Despite the fact that these children are initially raised as girls, the majority of them (60%) will eventually choose to live as heterosexual males, apparently due to the *organizing* effect of testosterone on early brain development and the *activating* effect of testosterone in puberty (Swaab, 2004; Swaab and Garcia-Falgueras, 2009).

Boys who are born with a cloacal exstrophy, i.e., with bladder exstrophy and partial or complete absence of penis, are usually changed into girls immediately after birth. A survey showed that only 65% of these children who were changed into girls continued to live as girls once they had reached adulthood. Moreover, when individuals with gender dysphoria were excluded, this number dropped to 47% (Reiner and Gearhart, 2004; Meyer-Bahlburg, 2005). These examples show again that a direct effect of testosterone on a developing boys' brain and a lack of such an effect on a developing girls' brain are crucial for the development of gender identity and sexual orientation.

Genetics

It should be noted that, although sex hormones are very important for gender identity and sexual orientation, sexual differentiation of the brain is not caused by hormones alone. *Genes* play a key role beyond the regulation of sex hormones. *SRY* and *ZFY* (Zinc Finger Protein Y-Linked) are possible candidates for a direct action on sexual differentiation of the brain since they are expressed up to very advanced ages in the human brain, even though strictly speaking the role of these genes in sexual differentiation stops during development (Mayer et al., 1998; Swaab, 2003; Bocklandt and Vilain, 2007). *SRY* may also play a role in brain sexual differentiation by acting as an epigenetic regulator (Sekido, 2014; Rosenfeld, 2017).

Strong genetic components are also found in gender dysphoria. A twin study found a significant additive genetic component accounting for 62% of the variance

(Coolidge et al., 2002), and another study observed that siblings of transgender persons have a higher risk of becoming transgender (Gómez-Gil et al., 2011). In addition, the genetic background of gender dysphoria is supported by a case report of a mother and daughter who became father and son (Sajevets et al., 2015). Moreover, chromosomal abnormalities, polymorphisms, and functional variants of genes and alleles relating to sex hormones are associated with gender dysphoria (Ngun et al., 2011; Fernández et al., 2014, 2018; Foreman et al., 2018). A recent study investigated sex hormone-signaling genes in 380 transgender women and 344 control men found significant associations between gender dysphoria and ER α , steroid 5- α reductase 2 (SRD5A2), and steroid sulfatase (STS) alleles, as well as ER α and sulfotransferase (SULT2A1) genotypes, which unveils the polygenic basis of gender dysphoria. In addition, they found an overrepresentation of several allele combinations involving the AR in transgender women (Foreman et al., 2018). Furthermore, several studies have shown that longer CAG repeats in the AR are correlated with decreased transcriptional activity of this receptor, leading to impaired androgen signaling (Chamberlain et al., 1994; Kazemi-esfarjani et al., 1995; Irvine et al., 2000). A meta-analysis of five studies consisting of 795 transgender women and 1355 control men in total has found a significant association between longer CAG repeats in the AR and transgender women, indicating that impaired androgen signaling may be linked to gender dysphoria (D'Andrea et al., 2020), likely in combination with other genes (Foreman et al., 2018). The importance of the ER and AR in the development of gender dysphoria in transgender women has also been confirmed by the research of (Fernández et al., 2018). These functional variants of genes and alleles may contribute to feminization or undermasculinization of the brain.

Although one would presume that, from an evolutionary point of view, genes related to homosexuality would not be favorable for reproduction and would therefore dissipate from the gene pool. The research by Camperio-Ciani et al. (2004) may provide an explanation for this paradox: the maternal gene that is related to male homosexuality is also related to female fecundity (Camperio-Ciani et al., 2004; Ciani et al., 2009; Fisher et al., 2018). Female relatives of homosexual males appear to have higher fecundity compared to female relatives of heterosexuals. The corresponding regions of this sexually antagonistic phenomenon were presumed to reside on the X chromosome and an autosomal chromosome (Ciani et al., 2008), possibly chromosome Xq28 (Hamer et al., 1993; Hu et al., 1995; Sanders et al., 2015) and chromosome 8 (Sanders et al., 2015). Sanders et al. (2015) performed a large genome-wide scan containing 409 pairs of homosexual brothers and found a significant linkage between chromosome 8 and chromosome Xq28

associated with male sexual orientation (Sanders et al., 2015). Chromosome Xq28 contains genes for arginine vasopressin receptor 2 and cyclic nucleotide gated channel alpha 2, which mediate social behaviors in humans and sexual behaviors (and olfactory signal transduction) in mice, respectively, and can be relevant for male sexual orientation (Brunet et al., 1996; Ebstein et al., 2012; Wang et al., 2019). However, the study by Sanders et al. (2015) only identified the region linked to male sexual orientation, and some studies have shown conflicting results (Rice et al., 1999; Mustanski et al., 2005). Thus further investigation of the specific genes related to this complex trait is certainly warranted.

Recently, a large GWAS study was performed on 477,522 individuals and replicated in 1524 individuals, in which genetic variants relating to same-sex sexual behavior were analyzed (Ganna et al., 2019). This study demonstrates the high complexity and heterogeneity of same-sex sexual behavior. Genome-wide, the study has identified many loci underlying same-sex sexual behavior, of which five autosomal loci showed significant association. One of the loci is associated with male pattern balding and was thought to be involved in sex-hormone regulation, while another locus is strongly linked to olfaction. Sex-hormone regulation and olfaction are overlapping processes and are related to sexual development and sexual preference (Valdes-Socin et al., 2014). Previous research has shown that olfactory processing of pheromones is dependent on sexual orientation (Savic et al., 2005; Berglund et al., 2006). Interestingly, there were other traits associated with same-sex sexual behavior, such as the personality trait “openness to experience,” risky behavior, and mental disorders (Ganna et al., 2019). The findings of this study reject the binary phenotype for sexual orientation and emphasize the use of separate measures of attraction for different sexes when analyzing sexual orientation in order to include higher diversities (e.g., bisexual, a-sexual).

Epigenetics

Epigenetic changes, such as acetylation and methylation in genes, are also found to affect the development of sex differences in rodent brain and behavior (McCarthy et al., 2009; McCarthy and Nugent, 2015; Szyf, 2019; M. Szyf, unpublished data) and are involved in the regulation of social, maternal, and sexual behavior (McCarthy and Nugent, 2015; Forger, 2016). The process by which early exposure to gonadal hormones organizes the brain and creates the “cellular memory” elicits a sexual dimorphic response in adulthood. This process is also thought to occur through epigenetic modifications (McCarthy et al., 2009; Forger, 2016). Gene expression is modulated through changes in chromatin, resulting in new cellular

phenotypes with the same genotype. Epigenetic changes found in the germ line are inherited, while those in somatic cells generally persist only for one lifetime and are largely context dependent (Crews, 2008), i.e., depending on different activities in the genome, the environment, drug exposure, or social experience (McCarthy and Nugent, 2013). In the nervous system, these modifications regulate the formation of neurons and neuronal apoptosis (Lv et al., 2013), and they control synaptic plasticity and may therefore be involved in memory and learning (Guan et al., 2015). A series of observations shows that ARs and ERs interact in rodents with histone modifying enzymes, which are associated with neural sexual differentiation (Tsai et al., 2009; Matsuda et al., 2012; McCarthy and Nugent, 2013; Zhang et al., 2013). The human central nucleus of the bed nucleus of the stria terminalis (BSTc) is sexually dimorphic, with a volume and number of cells that is two times larger in males as compared to females (Zhou et al., 1995a; Kruijver et al., 2000). Inhibition of histone deacetylases (HDACs) with valproic acid prevented the increased Histone 3 (H3) acetylation level during the critical period of brain sexual differentiation and masculinization of the BSTc in mice (Murray et al., 2009). Intraventricular injection of HDACs inhibitors also inhibited male sexual behavior (McCarthy and Nugent, 2013). Shen et al. (2015) found sex differences in Histone 3-Lysine-4 trimethylation (H3K4me3) in 248 genes and loci in the BSTc and preoptic area (POA) of mice, and those H3K4me3s that increased in females were associated with synaptic function (Shen et al., 2015). DNA methylation is necessary for cellular differentiation, suppression of repetitive elements, imprinting, and inactivation of the X chromosome (Vialou et al., 2013). Reduction of expression of proteins attracted by methylated cytosine that interacts with HDACs in the amygdala of neonatal rats reduced social playing behavior in males and eliminated the usual sex difference in this behavior (Kurian et al., 2008). Female rats have more DNA methyltransferase activity and more methylation sites in the POA than males. These sex differences were eliminated when the female rats were treated with estradiol after which their sexual behavior masculinized, suggesting that brain feminization was maintained by suppressing masculinization through DNA methylation (Nugent et al., 2015).

Endocrine disruptors

A relatively new field is that of endocrine disruptors, i.e., compounds present in the environment that pass the placenta and intervene in the interaction between endogenous sex hormones and the developing brain. This concerns, e.g., some pesticides, herbicides, plastic

contaminants (such as phthalates), pharmaceuticals (such as diethylstilbestrol, DES), and dietary components (such as phytoestrogens). Recent data give rise to considerable concerns about such effects during critical periods of human development. For example, endocrine disruptors may contribute to the earlier onset and longer duration of puberty in girls (Lee et al., 2019). In addition, the chemicals used in plastics can have a detrimental effect on human brain development and sexual differentiation. Due to endocrine disruption and toxicity, the widely used Bisphenol-A (BPA), which is an important precursor of plastics, has been replaced by its analogues Bisphenol-B (BPB) and Bisphenol-F (BPF) (Usman et al., 2019). However, several studies show that BPB has even an increased toxicity compared to BPA (Chen et al., 2002; Rosenmai et al., 2014; Russo et al., 2018), in addition to its estrogenic and antiandrogenic effects (Kitamura et al., 2005; Usman and Ahmad, 2016). Likewise, BPF has estrogenic, progestogenic, and antiandrogenic properties and has as much influence on hormone regulation as BPA (Rochester and Bolden, 2015; Roelofs et al., 2015; Feng et al., 2016; Mesnage et al., 2017; Lei et al., 2018a,b; Usman et al., 2019). In zebrafish, BPF exposure caused developmental toxicity by inducing aromatase activity, disrupting sexual differentiation and reproductive functions (e.g., less eggs and hatching), and affecting 5 α -reductase (Yang et al., 2017, 2018a,b). Humans are extensively exposed to these compounds, and we showed their presence in human post-mortem brain (Van Der Meer et al., 2017; Charisiadis et al., 2018). This can have a considerable impact on the endocrine systems and may, therefore, also influence sexual differentiation of the fetal brain during pregnancy. Despite these reports, the use of BPB and BPF are expected to increase in the future (Usman et al., 2019).

Prenatal exposure to phthalates, the antiandrogenic weakeners in plastics, is associated with less male-typical behavior in boys (Swan et al., 2010). In addition, intrauterine exposure to mild analgesics, which are antiandrogenic prostaglandin inhibitors, is a risk factor for the development of male reproductive disorders (Kristensen et al., 2011). Moreover, boys whose mother had been exposed to pesticides during pregnancy had smaller testicular volume and penile length at the ages between 6 and 11, while girls showed earlier breast development (Wohlfahrt-Veje et al., 2012a,b). What these observations may mean for later gender identity and sexual orientation for such children should be investigated.

Between 1939 and 1960, around two million expectant mothers in the United States of America and Europe were prescribed the synthetic estrogen known as DES, in the belief that it would prevent miscarriages. (It did not, in fact, but doctors like to prescribe medicines, and patients are always keen to be treated.) DES turned out

to increase the likelihood of bisexuality and homosexuality in the daughters of women given this drug (Ehrhardt et al., 1985). In addition, a high prevalence of gender identity problems has been reported in DES sons (Kerlin, 2005). A formal study of these alarming data is certainly warranted. Moreover, in a relatively small group of patients who were prescribed medication, taken during pregnancy, that inhibits the breakdown of sex hormones, i.e., old anti-epileptics such as diphantoin or phenobarbital, a high prevalence of gender dysphoria was observed in their children (Dessens et al., 1999).

Prebirth exposure to nicotine also increases the likelihood of lesbian daughters. Pregnant women suffering from serious stress are also more likely to give birth to homosexual children, because their raised levels of the stress hormone cortisol affect the production of fetal sex hormones (Ellis and Cole-Harding, 2001).

Immune response

It was found that the more older brothers a boy has, the greater the chance that he will develop gender dysphoria (Gómez-Gil et al., 2011). Each older brother also increases the odds of male homosexuality by approximately 33%. A recent study by Bogaert et al. (2017) has found an association between sexual orientation in male offspring and maternal immunoreactivity against the protein neuroligin 4 Y-linked (NLGN4Y; isoforms 1 and 2) (Bogaert et al., 2017). NLGN4Y plays a role in male fetal brain development and is primarily expressed on the extracellular surface—thus, it is exposed to maternal antibodies. In the plasma from women with homosexual sons (with or without older brothers), higher levels of anti-NLGN4Y were found compared to the plasma of women with heterosexual sons or no sons. This finding possibly elucidates this “fraternal birth order effect” and provides more insight into the development of male sexual orientation.

Self-organization

Lastly, the process of *self-organization* of the human brain is also responsible for the fact that even identical twins often differ in their gender identity and sexual orientation. Self-organization makes each brain and neuronal system unique, even if the genetic background is the same, as is the case in identical twins (Swaab, 2019). Self-organization occurs in complex systems and its principles are reflected everywhere, e.g., in anthills, in business, and in the universe. Self-organization may even lead to the functioning of a population as one unit, or a superorganism. A good example of self-organization is the swarms of starlings or a school of fish.

Our brains are too complex to develop solely on the basis of genetic information, or to operate on the basis

of guidance by a single brain region. The major ingredients for self-organization are indeed present in the brain: (i) the network of brain cells is extremely complex and (ii) rapid communication between the brain components is possible, (iii) based on experience, also learning takes place in the local networks, (iv) as many processes as possible are delegated to a lower level, so that locally, automatically, and unconsciously, processes are controlled and decisions are made, and (v) there is no brain area that constantly follows and controls in detail local processes. The brain develops as a complex, self-organizing system, and this way it also continues to function throughout our lives (Singer, 1986). This means that, while developing, the system is looking locally for the best contacts. If cells have an intensive functional contact with each other that contact is strengthened. Cells that fire together strengthen their mutual connections. “Cells that fire together wire together” as Hebb (1949) summarized this mechanism concisely. If the contact is weak, it will disappear, and subsequently the corresponding brain cell will also disappear. Therefore cell death is a normal process during brain development. We produced five times more brain cells than we are finally left with; thus this is a mechanism of survival of the fittest. This process, which is what makes each brain different, is therefore also called “Neuronal Darwinism.” It is easy to understand that this process is involved in the variation of our gender identity and sexual orientation.

Social factors?

Although it is frequently assumed that brain development after birth also has an important influence on gender identity or sexual orientation, there is no proof of this assumption whatsoever. On the contrary, all the data indicate that both gender identity and sexual orientation arise in the womb. There is no evidence for the misconception that homosexuality is a “lifestyle choice” or it stems from social learning. Children of lesbian mothers and gay fathers are not more likely to become homosexual and are not different from children raised by heterosexual parents in terms of personality development, psychological development, sexual orientation, and gender identity (Greenfeld, 2005). Recently there is attention for the concept of sexual fluidity that was first raised by Kinsey et al. (2003). Sexual orientation reparative therapies, as conducted in history, were in fact based on this assumption. They were not effective, and probably doing harm to such individuals (Haldeman, 1994; Dehlin et al., 2015). Women were found to be more sexual fluid than men (Kinnish et al., 2005; Mock and Eibach, 2012; Breedlove, 2017; Safron et al., 2020). It should be noted that women are twice more likely to report a bisexual than an exclusively same-sex orientation, while men show the opposite pattern (Bailey et al., 2016). Indeed,

the bisexual category is the most unstable in terms of sexual orientation (Savin-Williams et al., 2012). There is, however, no evidence that social factors also play a role in sexual orientation.

DIFFERENCES IN THE HYPOTHALAMUS IN RELATION TO SEX AND GENDER DYSPHORIA

Permanent structural and functional differences in the brain resulting from the interaction of developing brain cells with genes, sex hormones, and other chemical compounds are the basis of sex differences in reproduction, gender role (behaving as a man or a woman in society), gender identity, and sexual orientation.

Our group was the first to find a structural sex difference in the human hypothalamus, more precisely in the sexually dimorphic nucleus of the preoptic area (SDN-POA) (Swaab and Fliers, 1985), later also called the interstitial nucleus of the anterior hypothalamus-1 (INAH1) (Allen et al., 1989), or the intermediate nucleus (InM) (Koutcherov et al., 2007). The number of cells in the SDN-POA peaks at the age of 2–4 years both in males and females, after which there is a decline in the number of cells in female which manifests the sexual differentiation. The volume of the nucleus and the number of cells decrease with age in a sexually dimorphic way, i.e., the decline occurs in the females and not in the males (Swaab and Hofman, 1988) (Fig. 31.1). Later, we showed that the number of galanin neurons in this nucleus was larger in men than in women (Garcia-Falgueras et al., 2011). A sex difference was also found in the number of neurons expressing vasoactive intestinal polypeptide in the suprachiasmatic nucleus (SCN), which appeared to be age dependent (Zhou et al., 1995b). In addition, a clear age-dependent sex difference was observed in the activity of the vasopressin (AVP) neurons of the supraoptic nucleus and in sex hormone receptor distribution in the hypothalamus and adjacent areas (Swaab, 2003). Moreover, men have a significantly larger number of corticotropin-releasing hormone (CRH) neurons in the hypothalamic paraventricular nucleus than women, and the total number of CRH neurons increases significantly with age in male control subjects, but is not seen in female control subjects (Bao and Swaab, 2007). Also in the accelerated aging procedure, i.e., in Alzheimer’s disease (AD), which shows pronounced changes in neurofibrillary proteins in the infundibular nucleus and posterior median eminence, neurofibrillary pathology was found in 79% of males over 60 years of age, but in only 6% of females (Schultz et al., 1996). In contrast, we observed more hyperphosphorylated Tau in the nucleus basalis of Meynert of female than of male AD patients (Salehi et al., 1998).

Subsequently a large number of structural and functional sex differences have been described in the adult

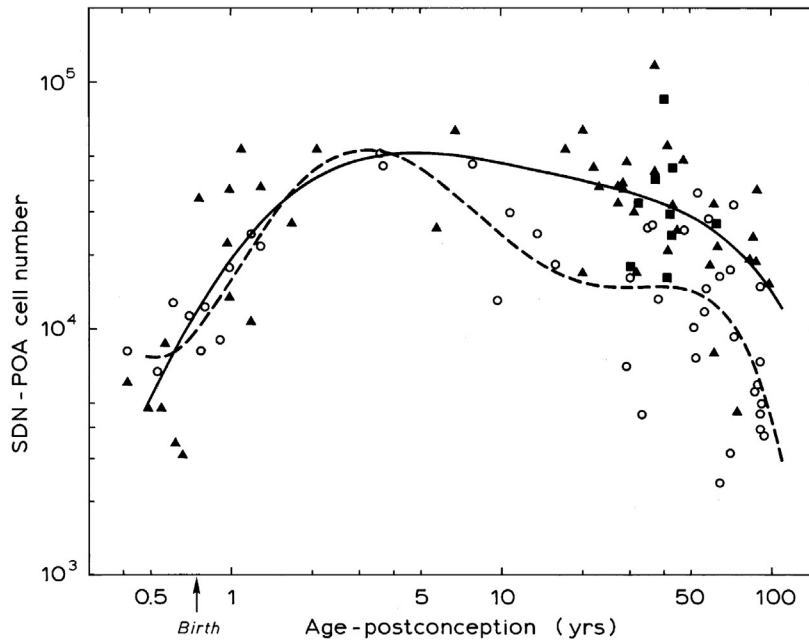


Fig. 31.1. Developmental and sexual differentiation of the human sexually dimorphic nucleus of the preoptic area (SDN-POA) of the hypothalamus in 99 subjects, log–log scale. Note that at the moment of birth the SDN-POA is equally small in boys ($\blacktriangle$) and girls ($\circ$) and contains about 20% of the cell number found at 2–4 years of age. Cell numbers reach a peak value around 2–4 years postnatally, after which a sexual differentiation occurs in the SDN due to a decrease in cell number in the SDN of women, whereas the cell number in men remains approximately unchanged up to the age of 50 years. The SDN-POA cell number in homosexual men ($\blacksquare$) does not differ from that in the male reference group ($\blacktriangle$). The curves are quintic polynomial functions fitted to the original data for males (continuous line) and females (dashed line). Adapted with permission from Swaab, D.F., Hofman, M.A., 1988. Sexual differentiation of the human hypothalamus: ontogeny of the sexually dimorphic nucleus of the preoptic area. *Brain Res Dev Brain Res* 44, 314–318.

human brain. For instance, a clear functional sex difference was observed in hypothalamic glucose metabolism that was higher in women than in men, as measured by PET (Kawachi et al., 2002). Several studies even successfully classified the sex of cisgender individuals with high accuracy by looking at structural data using different approaches (Menon and Krishnamurthy, 2019; Baldinger-Melich et al., 2020; Flint et al., 2020; Weis et al., 2020). Interestingly, the prediction accuracy is significantly decreased in transgender individuals, indicating that the transgender brain is indeed different from the cisgender brain (Baldinger-Melich et al., 2020; Flint et al., 2020). However, studies on sex differences are often inconsistent. A possible explanation for this is that most studies focus on mean differences instead of variance. A study by Wierenga et al. (2018) showed that males have higher variability (at both extremities) than females in cerebral white matter, hippocampus, cerebellar cortex, pallidum, putamen, cerebral cortex, but not in cortical thickness (Wierenga et al., 2018). Possibly, the presence of only one X-chromosome plays a role in increasing the variability in these brain structures and may also be associated with a greater risk for neurodevelopmental disorders. However, greater variance of brain regions in males is not linked to behavioral outcomes

(working memory, reading comprehension, and fluency), which may be more closely associated with experience (Wierenga et al., 2019). One of the reasons for inconsistent results regarding sex differences is that although there are sex differences for the *group* males and females, the *individual brain* does not belong in all aspects to one distinct category. There is a great variety in each brain in terms of the degree of masculinization or feminization of different brain areas. Rather, each brain is a unique mosaic of male and female characteristics (Joel et al., 2015, 2020). We confirmed in a collaborative study the presence of brain mosaicism for a number of hypothalamic nuclei (Joel et al., 2020).

Hypothalamic differences in relation to gender identity

Some brain differences seem to be related to gender identity. In 1995 in a postmortem study of human donor brains, we found a reversed sex difference in the BSTc, a structure that is known from animal experiments to be involved in sexual behavior. In men, the BSTc volume was twice as large as in women and contained twice as many somatostatin neurons (Zhou et al., 1995a; Kruijver et al., 2000) (Fig. 31.2). The same was true for

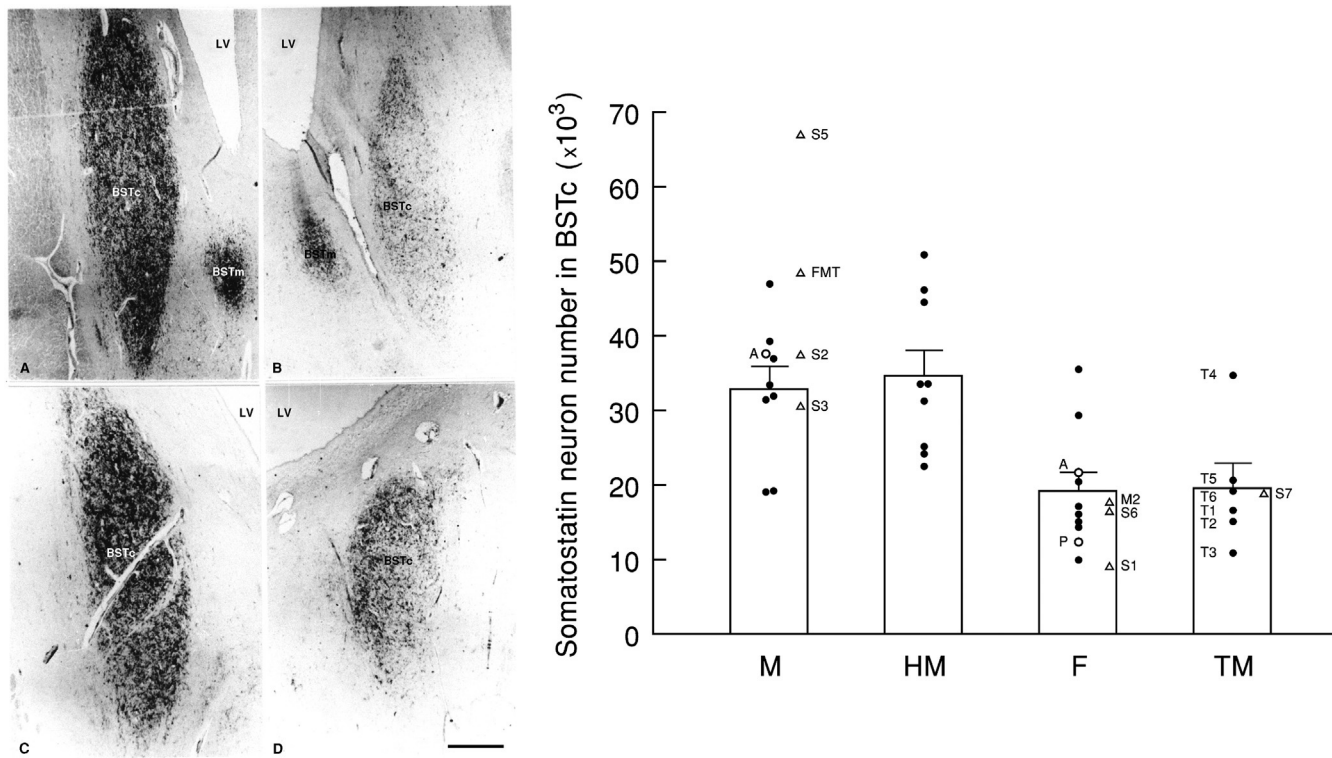


Fig. 31.2. *Left:* Representative sections of the central nucleus of the bed nucleus of the stria terminalis (BSTc) innervated by vasoactive intestinal polypeptide (VIP). (A) Heterosexual man; (B) heterosexual woman; (C) homosexual man; (D) male-to-female transgender. Scale bar, 0.5 mm. LV, lateral ventricle. Note there are two parts of the BST in (A) and (B): small medial subdivision (BSTm) and large oval-sized central subdivision (BSTc). Note also the sex difference (A vs B) and the fact that the male-to-female transgender (D) has a female BSTc in size and type of innervation. *Right:* Distribution of the BSTc neuron numbers among the different groups according to sex, sexual orientation, and gender identity. M, heterosexual male reference group; HM, homosexual male group; F, female group; and TM, male-to-female transgenders. The sex hormone disorder patients S1-6 and M2 indicate that changes in sex hormone levels in adulthood do not change the neuron numbers of the BSTc. The difference between the M and the TM group ($P < 0.04$) also becomes statistically significant according to the sequential Bonferroni method if S2, S3, and S5 are included in the M group or if S7 is included in the TM group ($P \leq 0.01$). Note that the number of neurons of the female-to-male transgenders (FMT) is fully in the male range. Whether the transgenders were male oriented (T1, T6), female oriented (T2, T3, T5), or both (T4) did not have any relationship with the neuron number of the BSTc. The same holds true for heterosexual and homosexual men. This shows that the BSTc number of somatostatin neurons is not related to sexual orientation. A, AIDS patient. The BSTc number of neurons in the heterosexual man and woman with AIDS remained well within the corresponding reference group, so AIDS did not seem to affect the somatostatin neuron numbers in the BSTc. P, postmenopausal woman. S1 (♀ 46 year of age): adrenal cortex tumor for more than 1 year, causing high cortisol, androstenedione, and testosterone levels. S2 (♀ 31 year of age): feminizing adrenal tumor that induced high blood levels of estrogens. S3 (♂ 67 year of age): prostate carcinoma; orchiectomy 3 months before death. S5 (♂ 86 year of age): prostate carcinoma; prostatectomy; and antiandrogen treatment for the last 2 years. S6 (♀ 25 year of age): Turner syndrome (45, X0; ovarian hypoplasia). M2 (♀ 73 year of age): postmenopausal status. *Left panel:* With permission from Zhou JN, Hofman MA, Swaab DF (1995b). VIP neurons in the human SCN in relation to sex, age, and Alzheimer's disease. *Neurobiol Aging* 16: 571–576. *Right panel:* With permission from Kruijver, F.P.M., Zhou, J.N., Pool, C.W., et al., 2000. Male-to-female transsexuals have female neuron numbers in a limbic nucleus. *J Clin Endocrinol Metab* 85, 2034–2041.

the INAH-3, which was also found to be two times larger in men than in women and to contain twice as many neurons (Garcia-Falgueras and Swaab, 2008). In male-to-female (MtF) transgenders, the INAH-3 and BSTc were both found to have a female volume and cell number. The only female-to-male transgender person available to us for study so far had a BSTc and INAH3 with clear male characteristics (Fig. 31.2). These sex reversals were found not to be influenced by circulating hormone levels in adulthood and seem thus to have arisen during development. The BSTc appeared to be similar in size and cell number in heterosexual and homosexual men. In contrast, the INAH3 of homosexual men had a size similar to the INAH3 of females (LeVay, 1991), and this brain area is thus also related to sexual orientation.

Gonadotropin-releasing hormone (GnRH) regulates the onset of puberty, while GnRH is, in turn, regulated by kisspeptin and neurokinin B (NKB) signaling (Goodman et al., 2007; Topaloglu et al., 2009). NKB and kisspeptin signaling in the human INF were found to be sexually dimorphic in adulthood. Interestingly, this sex difference shows, as one of the few brain areas, female dominance, i.e., it is larger in women than in men. In addition, these studies showed that male-to-female transgenders display a female-typical NKB and kisspeptin immunoreactivity, in accordance with their gender identity. No difference was observed in the number of kisspeptin-immunoreactivity neurons in human Immunodeficiency virus (HIV) positive homosexual compared to HIV positive heterosexual men (Taziaux et al., 2012, 2016). Consequently, so far only the sex difference reversal in the BSTc and INF seems to be specifically in accordance with gender identity and not related to sexual orientation. There are also functional sex differences in human hypothalamus. Functional scanning studies have established that when humans detect pheromones, it leads to the activation of the anterior hypothalamus in a gender identity-dependent way (Savic et al., 2001, 2009; Berglund et al., 2008). The stimulation pattern for MtF transgenders fell between that of men and women (Berglund et al., 2008). In a functional MRI study, prepubescent children and adolescents with gender dysphoria were presented a male chemo-signal (androstadienone) to evoke hypothalamic activation (Burke et al., 2014). Adolescents with gender dysphoria showed a clear sex-atypical hypothalamic activation after olfactory stimulation with androstadienone. Prepubertal girls with gender dysphoria did not differ significantly from control girls or boys, while prepubertal boys with gender dysphoria generally responded more closely to their natal sex. An explanation for the discrepancy between prepubertal children and adolescents is the heterogeneity of the prepubertal group. According to biographical interviews, childhood gender dysphoria

does not always persist into adolescence, and the ages 10–13 are critical for the developmental trajectory of gender dysphoria (Steensma et al., 2011).

Hypothalamic differences in relation to sexual orientation

Around the early 1990s, several structures in the hypothalamus related to sexual orientation have been identified. Swaab and Hofman (1990) found that the volume and AVP-cell number of the SCN of homosexual males are twice as large as in heterosexual males (Swaab and Hofman, 1990). As mentioned before, LeVay (1991) found subsequently that the INAH3 is similar in size between homosexual males and heterosexual females, which is more than twice as small as in heterosexual males (LeVay, 1991). The INAH3 is suggested to be a homologue for the sexual dimorphic nucleus in the POA of the hypothalamus in rodents. In contrast, the anterior commissure, a tract connecting the left and right temporal cortex, is larger in homosexual men as compared to both heterosexual women and heterosexual men (Allen and Gorski, 1992).

Functional brain scanning has also revealed differences in relation to sexual orientation. The hypothalamus of homosexual men appeared to be less responsive to fluoxetine as that of heterosexual men, indicating different activities of the serotonergic system (Kinnunen et al., 2004). The influence of putative pheromones on sexual behavior has also been explored by PET. Pheromones are excreted in perspiration in concentrations 10 times higher in men than in women. While male pheromones stimulate the hypothalamus of heterosexual women and homosexual men in the same way, heterosexual men are not stimulated by a male scent. This indicates that pheromones may be a contributing factor in determining our choice of partner (Savic et al., 2005). Comparing heterosexual and lesbian women revealed that lesbians reacted to pheromones in a sex-atypical, almost reciprocal, way (Berglund et al., 2006, 2008).

CONCLUSIONS

Up to the 1980s it was thought that a child's behavior was made male or female by postnatal social influences. In the 1960s and 1970s this incorrect concept led to immediate operations of newborns with indeterminate sex organs, with the idea that the social influence would work best on gender identity when the environment was not in doubt. This misconception has ruined many lives due to ignorance of intrauterine sexual differentiation of the brain. The presence or absence of testosterone is a main factor causing the child to develop male or female sex organs between the 6th and 12th week of pregnancy, while the brain differentiates along male or female lines

in the second half of pregnancy. It is during that time that the feeling of being a man or a woman, i.e., our gender identity, and our sexual orientation, i.e., heterosexual, homosexual, or bisexual, are programmed into the hardware of our brains for the rest of our lives. This process of sexual differentiation of the brain is for the major part caused by a surge of testosterone in boys in the second half of pregnancy and the absence of such a peak in sex hormones in girls. In addition, sexual differentiation of the brain is influenced by genetic polymorphisms, epigenetic factors, and disturbed by medicines and chemicals in the environment. Moreover, the process of self-organization makes each brain unique. The result of all these factors involved in brain development in general and in sexual differentiation of the brain in particular is a great variability in all aspects of gender identity and sexual orientation. Variation has been the motor of evolution and will always remain present. The problem is, however, as Milton Diamond said: “*nature loves variability but society hates it.*” There is no evidence for postnatal social factors playing an essential role in the development of gender identity or sexual orientation. Sexual orientation reparative therapies were not effective and probably doing harm. Women were found to be more sexual fluid than men.

A large number of structural and functional sex differences in the brain have been described. Concerning gender dysphoria, there is the finding of a reversal in the BSTc volume and cell number of the sex difference in transgender people. This has, however, so far not resulted in a specific and sensitive biological marker for gender identity in the brain in early development *in vivo*. There is still need for such a biological marker for gender dysphoria, since the diagnosis is at present only based on a description of the person's own feelings.

ACKNOWLEDGEMENTS

A.-M. Bao is supported by the National Natural Science Foundation of China (91849125 and 81971268). D.F. Swaab's research is supported by the foundation Vrienden van het Herseninstituut.

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